Catalytic NH_3 decomposition over Ni/Al_2O_3 in a stagnation flow reactor

Sadaf Davari, Camilo Cárdenas, Matthias Hettel, Patrick Lott, Steffen Tischer, Sofia Angeli, Olaf Deutschmann, Karlsruhe Institute of Technology (KIT), Karlsruhe/ Germany

Introduction

Since ammonia (NH_3) decomposition has recently gained increasing interest because it is considered a viable chemical for the storage of green hydrogen, a nickel-based catalyst was used to investigate the catalytic decomposition of ammonia in a stagnation flow reactor (SFR) at different conditions. Spatially resolved concentration profiles, perpendicular to the catalytic surface, were obtained and compared with modeling data, highlighting the unique ability of the stagnation flow reactor to allow profile measurements within the boundary layer.

Material and Methods

To study the catalytic decomposition of ammonia in an SFR under various conditions, a 20 wt. %Ni/Al_2O_3 catalyst was prepared by means of incipient wetness impregnation and subsequently washcoated onto a ceramic stagnation disk using the spin spray coating technique. Different conditions were produced through changing the temperature, pressure, or NH_3 concentration in balance Ar at a total flowrate of 10 L min^{-1}. The NH_3 concentration profiles through the boundary layer were obtained by applying a capillary-based sampling technique, using a mass spectrometer for gas analysis [1]. The potential of the SFR in the research of catalytic reactions is highlighted by the numerical calculations using the DETCHEM software package [2] detailed microkinetic model developed by Appari et al.[3].

Results and Discussion

Fig. 1 shows the ammonia concentration profiles from light-off tests at a ramp rate of 3 °C min^{-1} up to 550°C, performed at different pressures, with all other parameters held constant. The results indicate that pressure variations have little effect on the decomposition reaction. The comparison of experiments with simulation data performed by the aforementioned reaction mechanism. The simulation predicts the catalytic behavior better at higher temperatures. At these temperatures, the reaction is limited by diffusion rather than kinetics due to the high conversion.

Fig. 2 shows spatially resolved experimental results obtained at varying pressures and ammonia inlet concentrations (while keeping all other parameters constant), in comparison with 1D simulations. Changing the NH_3 concentration or system pressure barely affects the boundary layer thickness, which is about 7 mm, and the ammonia conversion on the surface, which is around 95%. However, increasing the NH_3 inlet concentration (as shown in Fig. 1b) shifts the system to a kinetically controlled regime, reducing surface conversion to 85%.

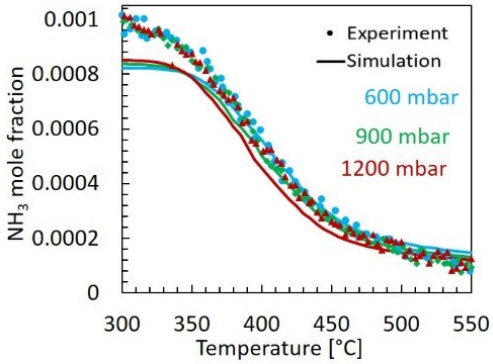

Figure 1. Comparison of the light-off measurements with simulation using Appari et al. [3] mechanism for 3 different system pressure values. NH_3 inlet mole fraction = 0.001, balance Ar, flowrate = 10 L min^{-1}, distance from catalyst surface = 0.5 mm.

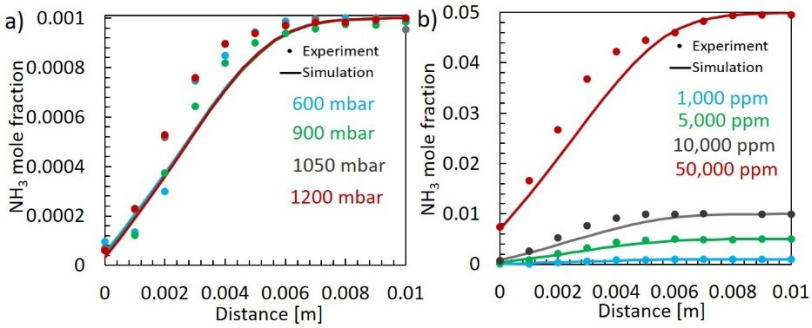

Figure 2. Comparison of experiment and simulation data of ammonia mole fractions through the boundary layer using the mechanism by Appari et al. [3] (NH_3 in balance Ar, flowrate = 10 L min^{-1}, T = 600 °C). **a)** NH_3 inlet mole fraction = 0.001, and p in mbar: 1) 600, 2) 900, 3) 1050, 4) 1200. **b)** p= 900 mbar, and NH_3 inlet in mole fractions [-]: 1) 0.001, 2) 0.005, 3) 0.01, 4) 0.05.

References

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